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Document

Architecture Overview

The DataFlow Platform follows a microservices architecture deployed on Kubernetes. The system is divided into four main layers:

[Data Sources] --> [Ingestion Layer] --> [Processing Layer] --> [Prese

Component Design

1. Ingestion Layer

- Apache Kafka for event streaming
- Debezium for CDC (Change Data Capture)
- Custom connectors for legacy systems

2. Processing Layer

- Apache Spark for batch processing
- Apache Flink for real-time streaming
- Data transformation using dbt

3. Data Storage

The platform uses a modern data lakehouse architecture:

- Raw Zone: S3-compatible storage (MinIO)
- Curated Zone: Delta Lake tables
- Serving Zone: PostgreSQL for dashboard queries

Important: All database schemas must follow

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4. Presentation Layer

- React-based dashboard
- GraphQL API gateway
- REST endpoints for legacy integration

Infrastructure Requirements

Component	Specification	Quantity
Kubernetes nodes	8 vCPU, 32GB RAM	6
Kafka brokers	4 vCPU, 16GB RAM	3
PostgreSQL	16 vCPU, 64GB RAM	2 (HA)
Object Storage	10TB	1 cluster

Integration Points

External API Gateway

The API Gateway handles all external communications. Rate limiting and authentication are enforced at this layer. Technical constraints and API limits are documented in: technical_constraints.md

Vendor Components

The real-time processing module depends on licensed components from StreamTech Solutions. Delivery and licensing terms are specified in vendor_agreement.md.

Security Architecture

All data flows are encrypted in transit (TLS 1.3) and at rest (AES-256).
For complete security requirements and compliance controls, refer to security_policy.md.



Deployment Strategy

- Phase 1: Core infrastructure + batch processing
 - Phase 2: Real-time streaming + API gateway
 - Phase 3: Advanced analytics + ML pipeline
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